

SAMPLE DOCUMENT - FICTITIOUS DATA FOR DEMONSTRATION ONLY

## PSV Sizing Calculation - PSV-101

Calculation CAL-PSV-101 Rev 0, per API 520 Part I, critical vapour flow, Kd 0.975, Kb 1.0, Kc 1.0.

|                             |                       |
|-----------------------------|-----------------------|
| Tag No.                     | PSV-101               |
| Set pressure                | 14.5 barg             |
| Relieving temperature       | 160 °C                |
| Required relieving capacity | 18500 kg/h            |
| Coefficient of discharge Kd | 0.975                 |
| Calculated orifice area     | 2,650 mm <sup>2</sup> |
| Selected orifice            | P (4P6)               |
| Selected orifice area       | 4,116 mm <sup>2</sup> |
| Rated capacity              | 28,700 kg/h           |

Inlet pressure loss below 3 % of set pressure; built-up back pressure below 10 % of set pressure.